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SEAWAYS – MPSS – FINANCE

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DR/SG

10th November, 1989.

Mr. Barry Ryan,
Managing Director,
Seaways Engineering (UK) Ltd,
York House, 5A Lancaster Road,
Wimbledon Village,
LONDON.
SW19 5DA

Dear Barry,

Please find enclosed two copies of our report on the market potential and economic impact of the MPSS production vessel in the North Sea sectors. We intend to have half a dozen or so copies professionally copied and bound. You should receive them within a week of receiving the initial copies.

I look forward to discussing the reports with you and Craig in due course. I apologise for the delay in getting the report to you. We have been completely overrun with work all summer. We have now been able to take on another full-time economist, Mr. Russell Dandie (from Marathon), and are hoping that will make life easier for a while!

Best regards,

David Rose

David Rose

Enc.

The economics of a number of model field situations in the UKCS (which reflect potential floating production systems) are examined under conventional producing systems and that available under the Seaways MPSS concept. In all cases substantial improvements in post-tax returns to investors were achieved with the substantial platform structure costs available with the Seaways MPSS design. Table S.3 below summarises the comparative post-tax profits for five model oil/gas fields.

Table S.3

Comparative Post-Tax Returns at $20 per Barrel (constant real)
(All absolute financial figures in £ million, 1989 prices)

Field Type	Rec. Res. Oil (mmbbl)	Rec. Res. Gas (bcf)	Conventional Development Scheme NPV at 10%	Conventional Development Scheme IRR%	Seaways MPSS Scheme NPV at 10%	Seaways MPSS Scheme IRR%
Alba	250	-	243	21.3	290	25.5
Andrew	90	200	14	11.6	49	16.9
Hudson	50	-	20	13.3	64	19.2
Keith	59	-	-3	9.6	29	14.2
Machar	65	-	0.4	10.1	37	15.8

The substantial capital cost savings of the MPSS concept (principally platform structure costs) enhance the post-tax returns for investors considerably, and in some instances transform marginal fields into attractive investments.

THE AUTHORS

Alexander G. Kemp is currently Professor of Economics at the University of Aberdeen. He was formerly Lecturer, Senior Lecturer and Reader. He previously worked for Shell, the University of Strathclyde and the University of Nairobi. For many years he has specialised in his research in petroleum economics with special reference to licensing and taxation issues. He has published over 80 books and papers in this field, including Petroleum Rent Collection around the World, Institute for Research on Public Policy (Canada), 1988. He has been a consultant on petroleum contracts and legislation to a number of oil companies, Governments, the World Bank, the United Nations and the Commonwealth Secretariat. He is a specialist adviser to the UK House of Commons Select Committee on Energy and a member of the Secretary of State for Scotland's panel of Economic Consultants. He is Director of Aberdeen University Petroleum and Economic Consultants (AUPEC), a division of Aberdeen University Research and Industrial Services Ltd. AUPEC provides consultancy services in petroleum economics.

David Rose is Senior Consulting Economist with Aberdeen University Petroleum and Economic Consultants (AUPEC) and Honorary Research Fellow in Economics at the University of Aberdeen. He has spent nine years in research into the economics of oil with special reference to taxation issues and has published many papers, jointly with Professor Alexander G. Kemp, in this field. He specialises in the application of sophisticated computer modelling techniques to the economics of oil and gas taxation. His activities include detailed computer-aided project appraisal studies, economic forecasting and impact studies, and market surveys for governments, public and private sector clients.

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SUMMARY

1. Objectives

Seaways Engineering (UK) Limited have designed a new innovative floating production system which is anticipated to cost a fraction of a conventional purpose-built semi-submersible. The objectives of this report are to identify those discoveries which could feasibly be exploited with the Seaways MPSS (Multi-Purpose Semi-Submersible) and to measure the economic benefits of the new floating production system for investors.

2. Methodology

The UK, Norwegian, Danish and Dutch sectors of the North Sea are all relatively mature oil and gas producing provinces. The increasing number of undeveloped discoveries - which are typically smaller and more technically difficult to exploit - are posing an increasing challenge for greater innovation and ingenuity from the offshore industry. The recent collapse in world oil prices has imposed a new economic environment where cost-effective technology is increasingly important.

In many cases the North Sea Governments have responded to the lower oil price environment with more progressive petroleum tax terms. To measure the economic impact of Seaways MPSS on new field developments in each of the North Sea sectors detailed computerised financial simulation models have been employed. The computer models

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